Xinyu (Catherine) Tian

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RESEARCH INTEREST — *Human data as a design and control signal for robot learning*

EDUCATION

Sep 2023 – Jun 2027

Tsinghua University Beijing, China

B.S., Zhili College · GPA 3.9 / 4.0, rank 3 / 35

- Advisors: Prof. Xiaorong Gao, Prof. Bo Hong
- Courses: Signals & Systems · Machine Learning & Pattern Recognition · Deep Learning · Bioinformatics · Biophysics · Linear Algebra · Probability & Statistics

2026 – present

University of California San Diego La Jolla, CA

Visiting Student

SKILLS

LANGUAGES English (IELTS 8.0) · Mandarin (native) · Arabic (elementary)

TECHNICAL Python · PyTorch · TransformerLens · VLM tooling · EEG & ECoG signal processing (FBCCA, TDCA, TRCA) · binocular stereo vision · 6D pose estimation · tensor decomposition · MATLAB / EEGLAB

RESEARCH

Vision–Language Model Interpretability

2026

Causal Alignment of Human Gaze and Vision–Language Model Attention

UC San Diego · GROUP LEADER

- Tested whether human fixation causally drives model attention in LLaVA and Qwen2-VL, patching activations at gaze-peak tokens against ground-truth fixations from the VQA-MHUG corpus
- Ablating gaze-peak tokens produced a significantly larger effect than matched random ablation (preliminary; $p < 0.01$, N = 120); the result is being re-run at higher power
- Implemented in TransformerLens, following the ARENA 3.0 mechanistic-interpretability methodology

Brain–Computer Interfaces for Real-Time Control

Feb 2025 – Jan 2026

Low-Fatigue Stimulus Design for SSVEP Spellers

Tsinghua University, Gao Lab · GROUP LEADER

- Replaced high-depth brightness flicker with low-depth and text-based stimuli, targeting the flicker sensation that limits how long a speller can be used rather than its peak accuracy
- Encoded three levels of Chinese linguistic structure — *zi* (character), *ci* (word) and *chengyu* (phrase) — to improve decoding robustness across units of different length
- Benchmarked FBCCA, TDCA and TRCA on accuracy and runtime across EEG from ten subjects; the low-depth study was conducted independently

Jun – Sep 2025

Real-Time Wheelchair Navigation via a Text-Based SSVEP Interface

Tsinghua University, Gao Lab · GROUP MEMBER

- Built a closed-loop interface issuing seven directional commands — forward, back, left, right, two rotations and stop — at 0.5 s decision intervals using calibrated TDCA
- Presented text-coded stimuli through an AR headset, trading peak decoding accuracy for stimuli a user can tolerate across a full session
- Validated and debugged the deployed system for the World Robot Contest

Computer Vision and Signal Processing for Clinical Assessment

Jan – Mar 2026

Automatic Detection of SPES Artifacts in ECoG

King's College London · GROUP LEADER

- Built an EEGLAB plugin that flags single-pulse electrical stimulation artifacts for seizure-onset-zone identification, replacing exhaustive visual inspection with a human-in-the-loop review step
- Across the five clinical recordings available, tuned detection caught every artifact with under 5% false positives and cut review time by roughly 80% (approximate figures on a small sample)
- Deliberately rule-based rather than learned, so a clinician can predict and audit its behaviour

Aug 2025 – present

Stereo Vision-Based ARAT Automation as Paralysis Rehabilitation Benchmark

Tsinghua University, Hong Lab · GROUP LEADER

- Designed the system architecture for scoring the Action Research Arm Test from stereo camera video, combining binocular stereo vision with CAD-based 6D pose estimation; implementation is ongoing
- Proposed tensor decomposition for dimensionality reduction in motor-imagery decoding
- Part of EBR, a minimally invasive BCI rehabilitation system funded as a key project by the Tsinghua Innovative Research Commission; supported ECoG data collection from patients

TALKS

Jun 2026

Active Use of Latent Tree-Structured Sentence Representation in Humans and Large Language Models

Journal club · Gao Lab Meeting

Nov 2025

NetFormer: An Interpretable Model for Recovering Dynamical Connectivity in Neuronal Population Dynamics

Journal club · Hong Lab Meeting

Oct 2025

Dream Engineering: From Targeted Memory Reactivation to Visual Reconstruction

Symposium · Tsinghua Academy Elite Scholars Program

Aug 2025

From Psychoacoustic Category Boundaries to Cortical Feature-Tuned Ensembles: Edward Chang's Research on Speech Perception

Journal club · Hong Lab Meeting

Mar 2025

Boolean Network Models: Toward Interpretable and Robust Biomodeling

Symposium · Tsinghua Academy Elite Scholars Program · with Zezhao Wu